**MECH 325**
**Homework Assignment #1**

**Problem 1 (Shigley Question 17-3 Modified)**

A 12 inch wide Polymide A-3 flat-belt drive is to connect a 5 inch diameter pulley to drive a larger pulley with a velocity ratio of 0.5. The center line distance is 20 feet. The small pulley turns at 1750 rpm as it delivers 3HP. Use a service factor $K_s = 1.25$ and design factor of 1.0.

a) Find $F_c$, $F_i$, $(F_1)_a$ and $F_2$ assuming operation at maximum tension limit
b) Find $H_a$, $n_{fs}$, and belt length
c) Find the dip

With what is available in Table 17-2 we will select the Polyamide A-3 belt with a thickness of 0.13 in. Also, let $b = 12$ in, $d = 5$ in with $n = 1750$ rev/min, $H_{nom} = 3$ hp, $C = 20(12) = 240$ in, velocity ratio = 0.5, $K_s = 1.25$, $n_d = 1$.

$$V = \pi d\, n / 12 = \pi(5)(1750) / 12 = 2290 \text{ ft/min}$$

$D = d$ / vel ratio = 5 / 0.5 = 10 in

Eq. (17-1): $$\theta_d = \pi - 2\sin^{-1}\frac{D-d}{2C} = \pi - 2\sin^{-1}\left[\frac{8-4}{2(216)}\right] = 3.123 \text{ rad}$$

Table 17-2: $t = 0.13$ in, $d_{min} = 4.3$ in, $F_a = 100$ lbf/in, $\gamma = 0.042$ lbf/in$^3$, $f = 0.8$

$$w = 12\,\gamma bt = 12(0.042)12(0.13) = 0.786 \text{ lbf/ft}$$

**(a)** Eq. (*e*), Sec. 17-2: $$F_c = \frac{w}{g}\left(\frac{V}{60}\right)^2 = \frac{0.586}{32.17}\left(\frac{1833}{60}\right)^2 = 17.0 \text{ lbf} \quad \textit{Ans.}$$

$$T = \frac{63\,025 H_{nom} K_s n_d}{n} = \frac{63\,025(2)(1.25)(1)}{1750} = 90.0 \text{ lbf}\cdot\text{in}$$

$$\Delta F = (F_1)_a - F_2 = \frac{2T}{d} = \frac{2(90.0)}{5} = 45.0 \text{ lbf}$$

Table 17-4: $C_p = 0.70$

Eq. (17-12): $(F_1)_a = bF_aC_pC_v = 12(100)(0.70)(1) = 840$ lbf *Ans.*

$$F_2 = (F_1)_a - [(F_1)_a - F_2] = 840 - 54 = 786 \text{ lbf} \quad \textit{Ans.}$$

Eq. (*i*), Sec. 17-2: $F_i = \dfrac{(F_1)_a + F_2}{2} - F_c = \dfrac{525.6 + 480.6}{2} - 17.0 = 486.1 \text{ lbf}$ *Ans.*

Eq. (17-9):

$$f' = \frac{1}{\theta_d}\ln\left[\frac{(F_1)_a - F_c}{F_2 - F_c}\right] = \frac{1}{3.123}\ln\left(\frac{525.6 - 17.0}{480.6 - 17.0}\right) = 0.0297$$

The developed friction is below the actual friction of 0.8 so the system is safe to use.

**(b)** The transmitted horsepower is, with $\Delta F = (F_1)_a - F_2$ =840-786 =54 lbf,

$$H = \frac{(\Delta F)V}{33\,000} = \frac{45(1833)}{33\,000} = 2.5 \text{ hp} \quad \textit{Ans.}$$

$$n_{fs} = \frac{H}{H_{\text{nom}} K_s} = \frac{2.5}{2(1.25)} = 1$$

Eq. (17-1): $$\theta_D = \pi + 2\sin^{-1}\frac{D - d}{2C} = \pi + 2\sin^{-1}\left[\frac{8 - 4}{2(216)}\right] = 3.160 \text{ rad}$$

Eq. (17-2): $$L = [4C^2 - (D - d)^2]^{1/2} + (D\theta_D + d\theta_d)/2$$

$$= [4(216)^2 - (8 - 4)^2]^{1/2} + [8(3.160) + 4(3.123)]/2 = 450.9 \text{ in} \quad \textit{Ans.}$$

**(c)** Eq. (17-13): $$\text{dip} = \frac{3C^2 w}{2F_i} = \frac{3(216/12)^2(0.586)}{2(486.1)} = 0.586 \text{ in} \quad \textit{Ans.}$$

$$\text{dip} = \frac{C^2 w}{96 F_i} = \frac{(240)^2(0.786)}{96(777.4)}$$

$$\approx 0.61 \text{ in}$$

1) 2023

**Problem 1 (Shigley Question 17-3)**

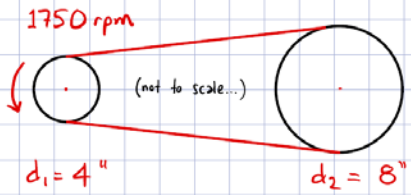
A 12 inch wide Polyamide F-1 flat-belt drive is to connect a 4 inch diameter pulley to drive a larger pulley with a velocity ratio of 0.5. The center line distance is 18 feet. The small pulley turns at 1750 rpm as it delivers 2HP. Use a service factor $K_s = 1.25$ and design factor of 1.0.

a) Find Fc, Fi, (F1)a and F2 assuming operation at maximum tension limit

b) Find Ha, safety factor nfs and belt length

c) Find the dip in the belt.

GIVEN: $b = 12"$, $d_1 = 4"$, $VR = 0.5$, $CD = 18\ ft = 216"$, $n_1 = 1750$ rpm, $H_{nom} = 2$ hp, $K_s = 1.25$, $n_d = 1.0$




$$VR = \frac{d_1}{D_2} = \frac{4"}{D_2} = 0.5 \rightarrow D_2 = 8"$$

$$n_2 = n_1 \cdot \frac{d_1}{D_2} = 1750\ rpm \cdot \frac{4"}{8"} = 875\ rpm$$

- assume operation @ maximum tension:

From TABLE 17-2 for F-1 Polyamide Belt:

$t = 0.05$ in
$F_a = 35$ lbf/in
$\gamma = 0.035$ lbf/in$^3$
$f = 0.5$

a) $F_c = \frac{w}{g}\left(\frac{V}{60}\right)^2$ (Eq. 17-e)

$$V = \pi dn/12\ [feet/min] = \pi(4\ in)\left(1750\ \frac{rev}{min}\right)\frac{1\ ft}{12\ in} = 1832.596\ ft/min.$$

$$w = 12\gamma bt\ \left[\frac{lbf}{ft}\right] = 12\ \frac{in}{ft} \cdot \left(0.035\ \frac{lbf}{in^3}\right)(12\ in)(0.05\ in) = 0.252\ lb/ft$$

$$F_c = \frac{0.252\ lb/ft}{32.17\ ft/s^2}\left(\frac{1832.596\ ft/min}{60\ s/min}\right)^2 = 7.30776\ lb$$

$$\boxed{F_c = 7.31\ lb}$$

$(F_1)_a = b \cdot F_a \cdot C_p \cdot C_v$ (Eq. 17-12)

From TABLE 17-4: $d = 4" \rightarrow C_p = 0.70$

$C_v = 1.0$ (see pg. 891, Shigley)

$$(F_1)_a = 12" \cdot \left(35\ \frac{lbf}{in}\right)(0.70)(1.0) = 294\ lb$$

$$\boxed{(F_1)_a = 294\ lb}$$

$$T = \frac{63\,025 \cdot H_{nom} \cdot K_s \cdot n_d}{n} \quad \text{(see pg. 889, Shigley)}$$

$$T = \frac{63025 \cdot 2\,hp\,(1.25)(1.0)}{1750\ rpm} = 90.036\ lb\cdot in$$

$$F_1 - F_2 = \frac{2T}{d} \quad \text{(Eq. 17-h)}$$

$$\hookrightarrow F_2 = F_1 - \frac{2T}{d} = 294\,lb - 2\left(\frac{90.036\ lb\,in}{4\ in}\right)$$

$$F_2 = 248.982\ lb \qquad \boxed{F_2 = 248.9\ lb}$$

$$F_i = \left(\frac{F_1 + F_2}{2}\right) - F_c \quad \text{(Eq. 17-i)}$$

$$= \frac{(294 + 249)}{2} - 7.31\ [lb] = 264.18\ lb \qquad \boxed{F_i = 264.2\ lb}$$

b)

$$H = \frac{(F_1 - F_2)\,V}{33\,000} \quad \text{(Eq. 17-j)} = \frac{(294 - 248.9)\,lb \cdot (1832.6\ fpm)}{33000\ \frac{ft\cdot lb}{min}/hp}$$

$$H_a = 2.50\ hp$$

$$n_{fs} = \frac{H_a}{H_{nom} \cdot K_s} \quad \text{(see pg. 889, Shigley)}$$

$$= \frac{2.50}{2(1.25)} = 1.0 \qquad \boxed{n_{fs} = 1}$$

- as expected, since it was assumed we are operating @ max tension

$$L = [4C^2 - (D-d)^2]^{1/2} + \frac{1}{2}(D\phi_D + d\phi_d) \quad \text{(Eq. 17-2)}$$

$$\begin{cases} \phi_d = \pi - 2\sin^{-1}\left(\frac{D-d}{2C}\right) \\ \phi_D = \pi + 2\sin^{-1}\left(\frac{D-d}{2C}\right) \end{cases} \quad \text{(Eq. 17-1)}$$

$$\phi_d = \pi - 2\sin^{-1}\left(\frac{8-4}{2(216)}\right) = 3.123\ rad$$

$$\phi_D = \pi + 2\sin^{-1}\left(\frac{8-4}{2(216)}\right) = 3.160\ rad$$

$$L = [4(216)^2 - (8-4)^2]^{1/2} + 1/2\left(8(3.16) + 4(3.123)\right) = 450.87$$

$$L = 450.9\ in$$

c) $$dip = \frac{C^2 w}{96 \cdot F_i} \quad \text{(Eq. 17-13)}$$

$$= \frac{(216\ in)^2 \cdot 0.252\ lb/ft}{96 \cdot (264.2\ lb)} = 0.4636''$$

$$dip = 0.464\ in$$

**Problem 2 and 3 (Mott Question 7-20 and 7-19)**

Using the information from Mott, design a V-belt drive to provide the required power and speed to the machinery in design problems 20 (Problem 2) and 19 (Problem 3a). Specify the belt type, the number of belts, the sheave sizes, the actual speed ratio, and the center line distance.

Problem 3b. Re-calculate Problem 19 using the information in the Gates manual. How does your solution compare to your solution in Problem 3a? Determine the required the belt pre-tension for the system and the minimum deflection at the midspan to confirm the required pre-tension.

**TABLE 7-27 V-Belt Drive Design Problems**

| Problem number | Driver type | Driven machine | Service (h/day) | Input speed (rpm) | Input power (hp) | Input power (kW) | Nominal output speed (rpm) |
|---|---|---|---|---|---|---|---|
| 18. | AC motor (HT) | Hammer mill | 8 | 870 | 25 | 18.6 | 310 |
| 19. | AC motor (NT) | Fan | 22 | 1750 | 5 | 3.73 | 725 |
| 20. | 6-cylinder engine | Heavy conveyor | 16 | 1500 | 40 | 29.8 | 550 |
| 21. | DC motor (compound) | Milling machine | 16 | 1250 | 20 | 14.9 | 695 |
| 22. | AC motor (HT) | Rock crusher | 8 | 870 | 100 | 74.6 | 625 |

*Note: NT indicates a normal-torque electric motor. HT indicates a high-torque electric motor.*

20. Design: Service Factor $= 1.4$; Design Power $= 1.4(40) = 56$ hp $\rightarrow$ 5V Belt

Ratio $= 1500/550 = 2.73$; $D_1 \approx 12(4000)/\pi(1500) = 10.2$ in

For $D_1 = 10.2$ in; $D_2 = 27.7$ in; $n_2 = 1500 \times 10.2/27.7 = 552$ rpm Ok

Rated Power $\cong 25.7$ hp by interpolation on Figure $7-15$ at 1500 rpm

$27.7 < CD < 114$; Use $CD \approx 36$ in; $L_p \approx 133.6$ in $\rightarrow$ Use $L_p = 132$ in

Actual $CD = 35.16$ in; $\theta_1 = 151.2°$; $\theta_2 = 208.8°$; $C_\theta = 0.92$; $C_L = 1.01$

Corrected Power $= (0.92)(1.01)(25.7) = 23.9$ hp/belt

Number of Belts $= 56/23.9 = 2.35 \rightarrow$ Use 3 Belts

Note: OK to use other reasonable values for CD (say 30, 40, 48 for example)

3)

**Problem 3a**

DriveR: AC Motor (Normal Torque) Speed = 1750rpm Input Power = 5 HP

DriveN: Fan Speed = 725 rpm (nominal) Service= 22 hours/day

**Problem 3b**

Re-calculate Problem 3a (Mott 7-19) using the information in the Gates manual (Gates VBelt Drive Design Manual). How does your solution compare to your solution in Problem 3a? Determine the required the belt pre-tension for the system and the minimum deflection at the midspan to confirm the required pre-tension.

a) GIVEN: $n_1 = 1750$ rpm $n_2 = 725$ rpm 22 hours/day

$H_{nom} = 5$ hp

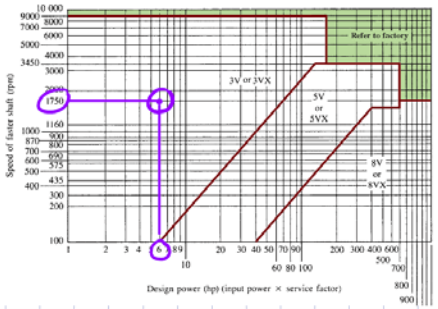
speed ratio $= \frac{1750}{725} = 2.414 \rightarrow D_2 = 2.414\, d_1$

$K_s = 1.2$ (Table 7-1) AC Motor, Fan, 22 hours/day

$H_d = K_s \cdot H_{nom} = 1.2\,(5\text{ hp}) = 6$ hp

Fig 7-13:

For 6 hp @ 1750 rpm, choose 3V belt(s)

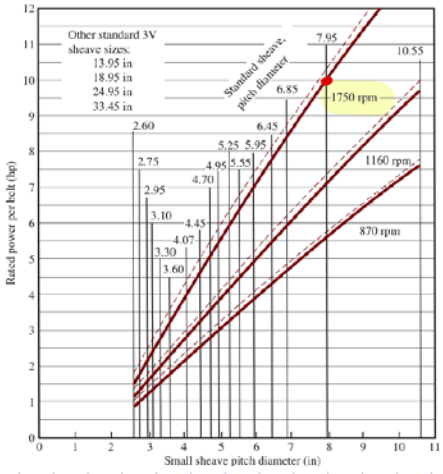



Target belt velocity $\approx$ 4000 ft/min. (see Mott, pg. 260)

$V_b = \frac{d_1 n_1}{2}$ $(v = r \cdot \omega)$

$d_1 = \frac{2V_b}{n} = \frac{2\left(4000\ \frac{ft}{min.}\right)}{\left(1750\ \frac{rev}{min}\cdot\frac{2\pi\ rad}{rev.}\right)} \cdot \frac{12\ in}{ft} \approx 8.731$ in

Fig. 7-14:




closest standard size $[d_1 = 7.95$ in$]$

| STANDARD $d_1$ (inches) | Approx. $D_2$ $(2.41 \cdot d_1)$ | NEAREST STANDARD $D_2$ (inches) | ACTUAL OUTPUT SPEED (rpm) |
|---|---|---|---|
| 6.85 | 16.51 | 18.95 | 632.6 |
| 7.95 | 19.16 | 18.95 | 734.2 |
| 10.55 | 25.43 | 24.95 | 739.9 |

Select $d_1 = 7.95$ in // $D_2 = 18.95$ in

- Actual output speed = 734.2 rpm (within 1.5% of nominal)

- Actual belt speed = $1750 \frac{r}{min.} \cdot \frac{2\pi \, rad}{rev} \cdot \frac{7.95 \, in.}{2} \cdot \frac{1 \, ft}{12} = 3642.3$ fpm ✓

$H_{tab}$ = Rated belt power = 10 hp (Fig. 7-14)

$\Delta P_R = 1.9$ (Fig. 7-17)

corrected $H_{tab}$ = 11.9 hp per belt - note: > Hd, so can likely use only 1 belt

CENTER DISTANCE $D_2 < CD < 3(D_2 + d_1)$ (Eq. 7-17)

$18.95 < CD < 50.7$

- start with 24 in. (low end to conserve space, since geometry not specified)

Length: $L = 2CD + 1.57(D_2 + d_1) + \frac{(D_2 - d_1)^2}{4CD}$ (Eq. 7-12)

$= 2(24) + 1.57(18.95 + 7.95) + \frac{(18.95 - 7.95)^2}{4(24)} = 91.49$ in

NEAREST STANDARD LENGTH = **90 in** (from Table 7-2)

Compute actual CD: (Eq. 7-13)

$B = 4L - 6.28(D_2 + d_1) = 4(90) - 6.28(18.95 + 7.95) = 191.07$

$CD = \frac{B + \sqrt{B^2 - 32(D_2 - d_1)^2}}{16}$

$= \frac{191.07 + \sqrt{(191.07)^2 - 32(18.95 - 7.95)^2}}{16} = 23.23$ in

**CD = 23.23 in**

Check wrap angle + correction factor:

$\theta_1 = 180° - 2\sin^{-1}\left[\frac{D_2 - d_1}{2CD}\right]$ $\quad \theta_2 = 180° + 2\sin^{-1}\left[\frac{D_2 - d_1}{2CD}\right]$

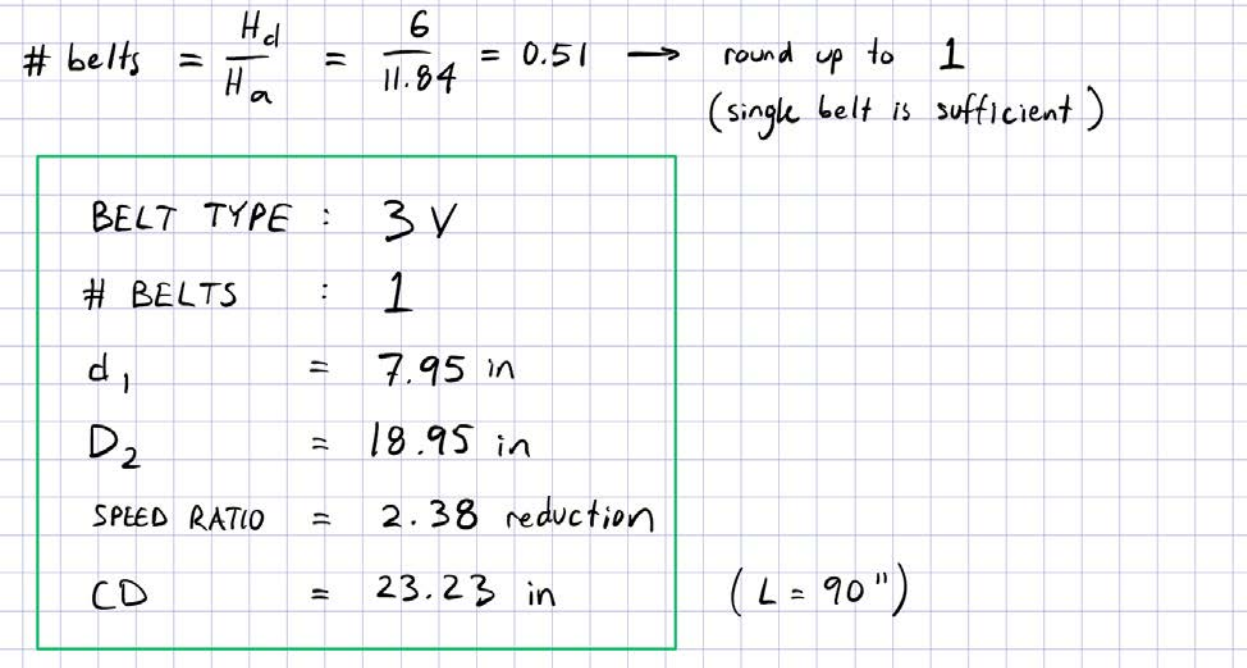
$\theta_1 = 152.61°$ $\quad \theta_2 = 207.4°$

From Fig. 7-18: $C_\theta = 0.93$

From Fig. 7-19: $C_L = 1.07$

$H_a = C_\theta C_L H_{tab} = (0.93)(1.07)(11.9)$ hp

**$H_a$ = 11.84 hp**




19. Design: Service Factor = 1.2; Design Power = 1.2(5) = 6.0 hp → 3V Belt

Ratio = 1750/72.5 hp = 2.41; $D_1 \approx 12(4000)/\pi(1750) = 8.7$ in

For $D_1 = 7.95$; $D_2 = 18.95$; $n_2 = 1750 \times 7.95/18.95 = 734$ rpm Ok

Rated Power ≅ 10.3 hp; $18.95 < CD < 80.7$; Try $CD \approx 30$ in

$L_p \approx 103$ in → Use $L_p = 100$ in; Actual $CD = 28.35$ in

$\theta_1 = 157.6°$; $\theta_2 = 202.4°$; $C_\theta = 0.94$; $C_L = 1.09$

Corrected Power = $(0.94)(1.09)(10.3) = 10.55$ hp/belt – One Belt Required

Note: OK to use other reasonable values for CD (say 20, 24, 36, 48 for example)

b)

**Problem 3b**

Re-calculate Problem 3a (Mott 7-19) using the information in the Gates manual (Gates VBelt Drive Design Manual). How does your solution compare to your solution in Problem 3a? Determine the required the belt pre-tension for the system and the minimum deflection at the midspan to confirm the required pre-tension.

Same scenario, using Gates Manual

GIVEN: $n_1 = 1750$ rpm  $n_2 = 725$ rpm  22 hours/day
$H_{nom} = 5$ hp

speed ratio $= \frac{1750}{725} = 2.414 \rightarrow D_2 = 2.414\, d_1$

GATES : Table No. B1

$K_s = 1.3 \rightarrow H_d = K_s \cdot H_{nom} = 1.3(5) = 6.5$ hp

$H_d = 6.5$ hp

Fig. No B1 : Select 3V belt(s)

Speed Ratio $\approx 2.41$

Table No. B2 : Min sheave diameter (6.5 hp @ 1750 rpm)
$\approx 3.0$ in

Table No. B6 : Select $d_1 = 3.35$" (larger than min. of 3.0)
$D_2 = 8.00$"

Choose CD = 10.8" → use 3V 400 Belt

hp correction = 0.9 (Table No. B6, Pg B16)

Table No. B9 (Pg. B56) @ 3.35" / 1750 rpm = 2.8 hp/belt = $H_{tab}$

$\Delta P_R = +0.36$

$H_a = 0.9(2.8 + 0.36)$

$H_a = 2.844$ hp

# belts $= \frac{H_d}{H_a} = \frac{6.5}{2.844} = 2.28 \rightarrow$ ROUND UP to 3 belts

| | | |
|---|---|---|
| BELT TYPE | : | 3V |
| # BELTS | : | 3 |
| $d_1$ | = | 3.35 in |
| $D_2$ | = | 8.00 in |
| SPEED RATIO | = | 2.39 reduction |
| CD | = | 10.8 in |

COMPARISON:

- Gates suggests 3 belts while Mott suggested only 1
- allowable power per belt is much lower
- Gates also leads to significantly smaller sheave diameters, and therefore slower belt velocity

Required Pre-tension

Formula No. D19 $T_{st} = 15\left(\frac{2.5 - K_\phi}{K_\phi}\right)\left(\frac{(\text{Motor hp})(10^3)}{(N)(V)}\right) + \frac{MV^2}{10^5}$

$V = \frac{1750}{(\text{rpm})} \cdot \frac{3.35 \text{ in}}{2} \cdot \frac{1 \text{ ft}}{12 \text{ in}} \cdot \frac{2\pi}{\text{rev}} = 1534.8$ fpm

$M = 0.46$ (Table No. D27)

$K_\phi = 0.94$ (Table No. D26)

$N = 3$ belts  $Y = 4$

$\rightarrow T_{st} = 24.894\left(\frac{5000}{3(1534.8)}\right) + \frac{0.46(1534.8)^2}{10^5}$

$T_{st} = 28.12$ lb per strand

Minimum deflection @ midspan $= \frac{1.4\, T_{st}}{2} + Y = 7.23$ lb to deflect

### Span Length, Two Wheel Drives

Belt span length is needed for the deflection method of measuring V-belt installation tension. Span length can be measured on the drive or measured from a scale layout of the drive.

For V or V-flat drives using only two wheels (no idlers) span length can be calculated from the following formula:

**Formula No. D35**

$$t = C\left[1 - 0.125\left(\frac{D-d}{C}\right)^2\right]$$

where: t = span length, inches
C = center distance, inches
D = large sheave or pulley diameter, inches
d = small sheave diameter, inches

NOTE: D and d are Outside Diameters for Super HC and Datum Diameters for Hi-Power II and Tri-Power Molded Notch V-Belts.

$$t = 10.8\left[1 - 0.125\left(\frac{8 - 3.35}{10.8}\right)^2\right]$$

$$\approx 10.54$$

$$t = \frac{10.5}{64} = 0.165''$$

---

**Problem 4 (Mott 7-43)**

Select a synchronous belt drive system for Question 19 above. For this application you can decrease the nominal output speed to 700 rpm.

---

As in Q3, AC input (normal torque) = 5HP @ 1750rpm. Service 22 hours/day
Output is a Fan @ 700rpm

Velocity Ratio (nominal) = 1750/700 = 2.5

From Table 7-8, Service Factor = 1.8
Design Power = 1.8*5 = 9.0 HP

From Figure 7-27, we can select an 8mm. pitch belt (1750rpm@ 9 HP)

From Table 7-7, options for VR = 2.5 are

1. 32 tooth DriveR and 80 tooth DriveN
2. 36 tooth DriveR and 90 tooth DriveN

From Table 7-9, we determine the HP rating for 30mm. belt width
Option 1 32 teeth @ 1750 rpm $HP_{rated}$ = 21.2 HP
Option 2 36 teeth @ 1750 rpm $HP_{rated}$ = 24.9 HP

Select Option 1 as it exceeds Design Power Rating

From Table 7-4, $PD_1$ = 3.208” and $PD_2$ = 8.020”
Using the V Belt rule for initial Centerline Distance estimate
$D_2 < CD < 3 (D_1 + D_2)$
8” < CD < 33”

From Table 7-7, select 115 teeth belt CL = 8.97 inches

Use Table 7-11 to determine belt correction factor, $C_L$ . For 115 teeth, $C_L$ = 1.0 so rated horsepower adjusted = $P_{adjusted} = P_{rated} * C_L = 21.2 * 1.0 = 21.2$ HP
Safety factor = $P_{allowable} / P_{design} = 21.2/9.0 = 2.36$

Belt speed = $\omega R = 0.5 * PD_1 * n_1 * 2\pi/60 = 0.5 * 3.208" * 1750 * 2\pi/60 = 1470$ feet/min
1470fpm< 3500 fpm so belt speed is acceptable

Specifications: Input: P32-8MGT-30
Output: P80-8GMT-30
Belt: 920 – 8MGT
CD = 8.97 inches
SF = 2.36

4) **Problem 4 (Mott 7-43)**

Select a synchronous belt drive system for Problem 3a (Mott7-19) above. For this application you can decrease the nominal output speed to 700 rpm.

DriveR: AC Motor (Normal Torque) Speed = 1750rpm Input Power = 5 HP

DriveN: Fan Speed = 700 rpm (nominal) Service = 22 hours/day

GIVEN: $n_1 = 1750$ rpm $n_2 = 700$ rpm 22 hours/day

$H_{nom} = 5$ hp

speed ratio $= \frac{1750}{700} = 2.5$ REDUCTION

Table 7-8: $K_s = 1.8$ (AC Motor Normal, Fan, 16-24 hours daily)

$H_d = K_s \cdot H_{nom} = 1.8\,(5\text{ hp}) = 9$ hp $\quad \boxed{H_d = 9 \text{ hp}}$

Fig. 7-27

@ 9 hp, 1750 rpm → select **8 mm pitch**

Table 7-7:

@ VR = 2.5, 2 options:

| DriveR | DriveN |
|---|---|
| 32 | 80 |
| 36 | 90 |

Select CD = 11.40 in

Belt: 1040-8MGT, P.L. 40.945, 130 Teeth

- no space restrictions given, but select smaller for more compact design

Table 7-9:

@ 32 teeth, 1750 rpm, 30 mm belt width: $H_{nom} = 21.2$ hp [$PD_1 = 3.208$ in]

Table 7-11:

Length correction factor = 1.00 (no change) $\quad H_a = H_{nom} = 21.2$

↳ 1 belt is sufficient to deliver $H_d = 9$ hp

Check belt velocity:

$$V = \frac{PD_1 \cdot \omega_1}{2} = \frac{3.208\text{ in}}{2} \cdot \frac{1\text{ ft}}{12\text{ in}} \cdot \frac{1750\text{ rev}}{\text{min}} \cdot \frac{2\pi\text{ rad}}{\text{rev}} = 1469.74 \text{ fpm}$$

$V = 1470$ feet/min. acceptable.

- below recommended max. of 3500 fpm (see pg. 272) ✓

DRIVER: 32 Teeth

DRIVEN: 80 Teeth

Speed ratio: 2.5

BELT: 1040-8MGT-30: 8 mm pitch, 30 mm width, 130 teeth (single belt) $\quad$ CD = 11.40 in

$n_{fs} \approx \frac{21.2}{9} = 2.36$

**Problem 5 (Mott 7-26 Modified)**

Roller chain is to be used in a hydraulic forklift truck to elevate the forks. If two strands support the load equally which size would you specify for a design load of 2.5 Tonnes? What will be your final safety factor?

26. Load on each chain $= W = 2500$ lb; $T.S. = 10\,W = 25\,000$ lb

**Use No. 120 chain $(1\ ^1/_2\ in\ Pitch)$; $T.S. = 34\,000$ lb** [Table 7-12]

**Problem 6 (Mott Question 7-40 Modified)**

Design a chain drive to provide the required power and speed to the machinery in design problem 40. Specify the chain size, the sizes and number of teeth in the sprockets, the number of chain links (pitches), and the center line distance.

**TABLE 7–28 Chain Drive Design Problems**

| Problem number | Driver type | Driven machine | Input speed (rpm) | Input power (hp) | Input power (kW) | Nominal output speed (rpm) |
|---|---|---|---|---|---|---|
| 38. | AC motor | Hammer mill | 310 | 25 | 18.6 | 160 |
| 39. | AC motor | Agitator | 750 | 5 | 3.73 | 325 |
| 40. | 6-cylinder engine | Heavy conveyor | 500 | 40 | 29.8 | 250 |
| 41. | Steam turbine | Centrifugal pump | 2200 | 20 | 14.9 | 775 |
| 42. | Hydraulic drive | Rock crusher | 625 | 100 | 74.6 | 225 |

| CHAIN DRIVE DESIGN | Application: | Heavy conveyor driven by a 6-cylinder Engine | |
|---|---|---|---|
| | *Problem 7-40* | | |
| ***Initial Input Data:*** | | | |
| Power input: | *40* | hp | |
| Service factor: | *1.4* | | Table 7-17 |
| Input speed: | *500* | rpm | |
| Desired output speed: | *250* | rpm | |
| ***Computed Data:*** | | | |
| Design power: | 56 | hp | |
| Design speed ratio: | 2.00 | | |
| ***Design Decisions-Chain Type and Teeth Numbers:*** | | | |
| Number of strands: | *3* | | 1 2 3 4 |
| Strand factor: | *2.5* | | 1.0 1.7 2.5 3.3 |
| Required power per strand: | 22.40 | hp | |
| Chain number: | *80* | | Tables 7-7, 7-8, or 7-9 |
| Chain pitch: | *1.00* | in | |
| Nunber of teeth-Driver sprocket: | *14* | | |
| Computed no. of teeth-Driven sprocket: | 28.00 | | |
| *Enter: Chosen number of teeth:* | *28* | | Check availability from vendor |
| ***Computed Data:*** | | | |
| Actual output speed: | 250.0 | rpm | |
| Pitch diameter-Driver sprocket: | 4.494 | in | |
| Pitch diameter-Driven sprocket: | 8.931 | in | |
| ***Center Distance, Chain Length and Angle of Wrap:*** | | | |
| *Enter: Nominal center distance:* | *36* | pitches | 30 to 50 pitches recommended |
| Computed nominal chain length: | 93.1 | pitches | |
| *Enter: Specified no. of pitches:* | *94* | pitches | Even number recommended |
| Actual chain length: | 94.00 | in | |
| Computed actual center distance: | 36.432 | pitches | |
| Actual center distance: | 36.432 | in | |
| Angle of wrap-Driver sprocket: | 173.0 | degrees | Should be greater than 120 degrees |
| Angle of wrap-Driven sprocket: | 187.0 | degrees | |

**Problem 7 (Shigley Question 17-27 Modified)**

A 700 rev/min 25-hp squirrel-cage induction motor is to drive a two-cylinder reciprocating pump, out-of-doors under a shed. The nominal pump speed is 140 rev/min. Select a suitable number of strands of Number 80 chain and sprocket sizes.

7) **Problem 7 (Shigley Question 17-27 Modified)**

A 700 rev/min 25-hp squirrel-cage induction motor is to drive a two-cylinder reciprocating pump, out-of-doors under a shed. The nominal pump speed is 140 rev/min. Select a suitable number of strands of Number 80 chain and sprocket sizes.

$H_{nom} = 25$ hp

$n_1 = 700$ rpm

$n_2 = 140$ rpm

CHAIN: No. 80

Need 5 to 1 reduction

$N_2 = 5 N_1$

$H_d = H_{nom} \cdot K_s \cdot n_d$ (Eq. 17-38)

- choose $n_d = 1.1$, $K_s = 1.5$
  - outdoors
  - 2 cylinder engine (rough - moderate shock loads)
  - likely on for long hours (6-15 h/day)

$H_d = 25 \text{ hp} \cdot 1.5 \cdot 1.1 = \boxed{41.25 \text{ hp}}$

choose $N_1 = 17$ teeth

↳ $K_1 = 1.0$ (Table 17-22)

$N_2 = 5(17) = 85$ teeth → choose 84 from standard sizes (Table 17-21)

Find Rated Horsepower:

Table 17-20: No 80 CHAIN, 700 rpm (pre-extreme), (TYPE B LUBRICATION - bath) → 31 hp rated capacity

$H_{tab} = 31$ hp

$H_a = K_1 K_2 H_{tab}$

$= (1.0)(K_2)\, 31 \text{ hp} \geq H_d$ (- allowable must be greater than design power)

$31 \cdot K_2 \geq 41.25$

Choose $K_2 = 1.7$ (2 strands)

Check $31 \cdot (1.7) \geq 41.25$ ?

$52.7 \geq 41.25$ ✓ YES

| | |
|---|---|
| SPROCKET SIZES: | 17 & 84 TEETH |
| # of STRANDS: | 2 |

Note: actual nominal speed $= 700 \times \frac{17}{84}$

$= 141.7$ rpm

(within 1.2 %)

---

**Problem 8 (Question 17-31)**

A 2000-ft mine hoist operates with a 72-in drum using 6 ×19 monitor-steel wire rope. The cage and load weigh 8000 lbf, and the cage is subjected to an acceleration of 2 ft/s$^2$ when starting.

(*a*) For a single-strand hoist how does the factor of safety $n_f = F_f/F_t$, neglecting bending, vary with the choice of rope diameter?

(*b*) For four supporting strands of wire rope attached to the cage, how does the factor of safety vary with the choice of rope diameter?

(c) Would you expect your chosen diameter in part (a) to change depending on the end fitting selected (say flemish eye splice versus wire rope clips)? For the wire rope clips specify the number of clips, torque required, and minimum tie back required.

**17-31** Given: 2000 ft lift, 72 in drum, 6 × 19 MS rope, cage and load 8000 lbf, accel. = 2 ft/s$^2$.

**(a)** Table 17-24: $(S_u)_{\text{nom}} = 106$ kpsi; $S_u = 240$ kpsi (plow steel); Fig. 17-21: $(p/S_u)10^6 = 0.0014$

Eq. (17-44):

$$F_f = \frac{(p/S_u)S_u dD}{2} = \frac{0.0014(240)d(72)}{2} = 12.1d \text{ kip}$$

Table 17-24: $wl = 1.6d^2\,2000(10^{-3}) = 3.2d^2$ kip

Eq. (17-47):

$$F_t = (W + wl)\left(1 + \frac{a}{g}\right)$$

$$= (8 + 3.2d^2)\left(1 + \frac{2}{32.2}\right)$$

$$= 8.5 + 3.4d^2 \text{ kip}$$

Note that bending is not included.

$$n_f = \frac{F_f}{F_t} = \frac{12.1d}{8.5 + 3.4d^2}$$

| $d$, in | $n_f$ | |
|---|---|---|
| 0.500 | 0.650 | |
| 1.000 | 1.020 | |
| 1.500 | 1.124 | |
| 1.625 | 1.125 | ← maximum $n_f$ *Ans.* |
| 1.750 | 1.120 | |
| 2.000 | 1.095 | |

**(b)** Try $m = 4$ strands

$$F_t = \left(\frac{8}{4} + 3.2d^2\right)\left(1 + \frac{2}{32.2}\right)$$
$$= 2.12 + 3.4d^2 \text{ kip}$$
$$F_f = 12.1d \text{ kip}$$
$$n_f = \frac{12.1d}{2.12 + 3.4d^2}$$

| $d$, in | $n_f$ | |
|---|---|---|
| 0.5000 | 2.037 | |
| 0.5625 | 2.130 | |
| 0.6520 | 2.193 | |
| 0.7500 | 2.250 | ← maximum $n_f$ *Ans.* |
| 0.8750 | 2.242 | |
| 1.0000 | 2.192 | |

Comparing tables, multiple ropes supporting the load increases the factor of safety, and reduces the corresponding wire rope diameter, a useful perspective.

Part c). Refer to the notes on Canvas regarding the end conditions. In part a) the selected wire rope diameter is 1.625 inches.

i) Flemish eye splice efficiency defined as 92.5% for diameter 1.625 inches.
ii) Clips give an efficiency of 80% is installed properly.

## EFFICIENCY RATINGS

| TERMINATIONS | EFFICIENCY RATING |
|---|---|
| WIRE ROPE SPLICE | |
| FLEMISH EYE SPLICE | 1/4" TO 1" DIAMETER AND SMALLER 95% |
| | 1 1/8" TO 2" DIAMETER 92.5% |
| | 2 1/8" & LARGER DIAMETER 90% |
| PRESSED SLEEVE | 1" DIAMETER AND SMALLER 95% |
| | 1 1/8" TO 2" DIAMETER 92.5% |
| | 2 1/8" & LARGER DIAMETER 90% |
| HAND TUCKED SPLICE | ONLY TO BE USED WHERE SLING ENDS ARE |
| | TERMINATED AND NOT ALLOWED TO ROTATE |
| | 1/4" DIAMETER - 90%, 5/16" DIAMETER - 89%, |
| | 3/8" DIAMETER - 88%, 7/16" DIAMETER - 87%, |
| | 1/2" DIAMETER - 86%, 5/8" DIAMETER - 84% |
| | 3/4" DIAMETER - 82%, 7/8" DIA. & GREATER - 80% |
| SWAGE SOCKET | OPEN OR CLOSED 100% |
| SPELTER SOCKET | OPEN OR CLOSED 100% |
| WEDGE SOCKET | 75-90% DEPENDING ON DESIGN (Check with Wesco) |
| CLIPS | 80% IF INSTALLED CORRECTLY |

$$n_f = \frac{F_f}{F_t} = \frac{12.1d}{8.5 + 3.4d^2}$$

We can reduce the allowable strength by the efficiency to get the new safety factor. This is the same as reducing the safety factor by the efficiency value.

i) Flemish eye splice: 0.925 (1.125) = $n_f$ = 1.04 (just acceptable)
ii) Clips: 0.80(1.125) = $n_f$ = 0.9 (unacceptable)

Note: For part b) solution. For wire rope diameter of 0.75 inches we have:

i) Flemish eye splice (95% efficient for d = 0.75 inches) – 0..95(2.25) = $n_f$ = 2.14.
ii) Clips: 0.80(2.25) = $n_f$ = 1.80

For 0.75 inch diameter rope, the required number of clips are 4 with a minimum inches of turnback and a minimum torque on the tightening nuts of 130 foot pounds.

# WIRE ROPE CLIPS

Wire rope clips are useful for field installations. Correct installation with the use of a torque wrench, the termination efficiency will be 80%. It is important that installation instructions are followed carefully and the table below outlining minimum clips to be used and torque to be applied is strictly adhered to.



Wire rope clip installation is as follows:
1. Turn back specified amount of rope (see table) from thimble or loop. Apply first clip one base width from dead end of rope. Apply U-bolt over dead end of wire rope - live end rests in saddle (Never saddle a dead horse!). Tighten nuts evenly, alternating from one nut to the other until reaching the recommended torque.
2. When two clips are required, apply the second clip as near the loop or thimble as possible. Tighten nuts evenly, alternating until reaching the recommended torque. When more than two clips are required, apply the second clip as near the loop or thimble as possible, turn nuts on second clip firmly, but do not tighten. Proceed to Step 3.
3. When three or more clips are required, space additional clips between first two - take up rope slack - tighten nuts on all clips, alternating from one nut to the other until reaching the recommended torque.
4. Apply an initial load equal to or greater than the loads expected in use. Inspect for proper spacing and retighten the nuts to the

Step 1



Step 2



Step 3



| Rope Sizes Inches | Minimum number of Clips | Amount of Rope to Turn Back in Inches | Torque in Foot Pounds |
|---|---|---|---|
| 1/8 | 2 | 3 1/4 | 4.5 |
| 3/16 | 2 | 3 3/4 | 7.5 |
| 1/4 | 2 | 4 3/4 | 15 |
| 5/16 | 2 | 5 1/4 | 30 |
| 3/8 | 2 | 6 1/2 | 45 |
| 7/16 | 2 | 7 | 65 |
| 1/2 | 3 | 11 1/2 | 65 |
| 9/16 | 3 | 12 | 95 |
| 5/8 | 3 | 12 | 95 |
| 3/4 | 4 | 18 | 130 |
| 7/8 | 4 | 19 | 225 |
| 1 | 5 | 26 | 225 |
| 1 1/8 | 6 | 34 | 225 |
| 1 1/4 | 7 | 44 | 360 |
| 1 3/8 | 7 | 44 | 360 |
| 1 1/2 | 8 | 54 | 360 |
| 1 5/8 | 8 | 58 | 430 |
| 1 3/4 | 8 | 61 | 590 |
| 2 | 8 | 71 | 750 |
| 2 1/4 | 8 | 73 | 750 |
| 2 1/2 | 9 | 84 | 750 |
| 2 3/4 | 10 | 100 | 750 |
| 3 | 10 | 106 | 1,200 |



8)

**Problem 8 (Shigley Question 17-31)**

A 2000-ft mine hoist operates with a 72-in drum using 6 ×19 monitor-steel wire rope. The cage and load weigh 8000 lbf, and the cage is subjected to an acceleration of 2 ft/s$^2$ when starting.

(*a*) For a single-strand hoist how does the factor of safety $n_f = F_f/F_t$, neglecting bending, vary with the choice of rope diameter?

(*b*) For four supporting strands of wire rope attached to the cage, how does the factor of safety vary with the choice of rope diameter?

GIVEN: 6 × 19 wire rope, monitor steel

2000 ft shaft
Drum: 72" ø
Load: 8000 lbf
$a$ = 2 ft/s$^2$

From Table 17-24

$w = 1.60\ d^2$
$D = 26d - 34\ d$
$E_r = 12$ Mpsi $= 12 \times 10^6$ psi
$S_u = 106$ kpsi (nominal)

Table 17-27

$A_m = 0.40\ d^2$
$D = 30d - 45\ d$ ↑ better

a) $m = 1$ (single strand) $\quad S_u = 240$ kpsi

$$F_f = \frac{(P/S_u)\ S_u\ d\ D}{2} \quad \text{(Eq. 17-44)}$$

$$F_t = \left(\frac{W}{m} + wl\right)\left(1 + \frac{a}{g}\right) \quad \text{(Eq. 17-47)}$$

$m$ = 1 (# of strands)
$W$ = 8000 lbf
$w = 1.60\ d^2$ lbf/ft
$l$ = 2000 ft max
$a$ = 2 ft/s$^2$ max
$g$ = 32.17 ft/s$^2$

From Fig 17-21:

choose $P/S_u = 1.4/1000 = 0.0014$

- for 1,000,000 bends to failure

$$F_f = \frac{0.0014\ (240 \times 10^3\ \text{psi})\ (72\ \text{in})\ d}{2}$$

$$F_f = 12096\ d\ \text{[lb]}$$

$$F_t = \left(\frac{8000\ \text{lbf}}{1} + 1.60\ d^2\ \frac{\text{lbf}}{\text{ft}} \cdot 2000\ \text{ft}\right)\left(1 + \frac{2}{32.17}\right)$$

$$F_t = 8497.36 + 3398.94\ d^2\ \text{[lb]}$$

- $F_f$ is linear with respect to diameter

$F_t$ is a quadratic with respect to diameter

$$n_f = \frac{F_f}{F_t} = \frac{12096\ d}{(8497.36 + 3398.94\ d^2)}$$

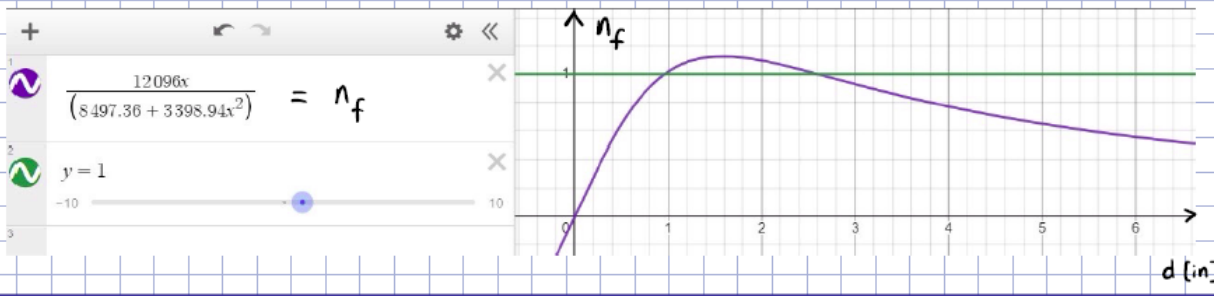
- factor of safety increases with d from 0 to maximum

of $n_{f\,max} = 1.125$ at $d = 1.581$ inches - standard $1\,1/2"$ would give best $n_f$

- However, below $d = 0.963"$, $n_f$ is below 1, indicating that the rope is not strong enough

- Similarly, above $d = 2.596"$, $n_f$ is below 1

  - the additional weight of the rope itself begins to overwhelm the added strength it provides at larger diameters




b) With 4 strands, $F_f$ remains unchanged:

$F_f = 12096\ d\ [lb]$ - fatigue on each strand is same as in Part a

$F_t$ is updated with $m = 4$

$$F_t = \left(\frac{8000\ lbf}{4} + 1.60\,d^2\,\frac{lbf}{ft^2}\cdot 2000\ ft\right)\left(1 + \frac{2}{32.17}\right) = 2124.34 + 3398.94\,d^2$$

$$n_f = \frac{F_f}{F_t} = \frac{12096\ d}{2124.34 + 3398.94\,d^2}$$

→ now, higher safety factors can be achieved at lower diameters, which was not possible with a single strand.

$n_{f\,max} = 2.251$ @ $d = 0.791$ in

$n_f \geq 1$ for d between 0.185 - 3.373 in

